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A visitor-enriched census in the U.S. cities using large-scale mobile positioning data

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Census data, as a traditional data source of resident socio-demographics, provides valuable information for decision-makers, researchers, and the public. While numerous efforts have been made to develop more comprehensive data products based on census datasets, most approaches treat census units as static and independent entities, overlooking their interactions. In this paper, we introduce the “visitor census” dataset, a semantically enriched census that incorporates human visitations extracted from large-scale mobile positioning data. We identified and validated the potential home locations of 3.58 million anonymous mobile phone users across seven U.S. metropolitan statistical areas in July 2021 and utilized home detection results to enrich the socio-demographic profile of the places users visited. The proposed data generation framework is adaptive, allowing future integration of diverse socio-demographic features at varying spatial and temporal scales. Overall, this visitor-based census represents an effort to enrich resident-based census knowledge by incorporating mobilities and spatial interactions in human digital traces, bridging the gap between aggregated and individual analysis, as well as between conventional census and mobile phone data.

Background & Summary

Census data, capturing a wide range of population and housing attributes, provides demographic and economic information at multiple spatial and temporal scales^{1,2}. In the United States, a comprehensive decennial census is supplemented by the American Community Survey (ACS), producing data at granular levels from the national scale down to the census block group³. Over the past several decades, both raw census datasets, exemplified by the Public Use Microdata Sample (PUMS)⁴ and harmonized datasets, such as the Integrated Public Use Microdata Series (IPUMS)⁵, have been made publicly accessible in electronic format. The increasing availability of both individual-level and aggregated data, driven by digital transformation, has enabled users to conveniently obtain tables and geographic files for a wide range of analytical applications⁶.

As the demand for more detailed, frequent, and high-quality census data grows, various initiatives have emerged to expand and refine existing datasets, significantly shaping contemporary socio-demographic research. These include the development of products such as LandScan⁷, the Longitudinal Tract Database (LTDB)⁸, WorldPop⁹, GHS-POP¹⁰, and Optimized Spatial Census Information Linked Across Time (OSCILAT)¹¹, alongside a proliferation of census data integration and visualization applications¹². Much of this work focuses on improving population estimates or calibrating long-term population data by leveraging complementary data sources¹³. Efforts to enrich the semantic content of census datasets are also mushrooming, as evidenced by the LEHD Origin-Destination Employment Statistics (LODES)¹⁴ and the CDC Social Vulnerability Index (SVI)¹⁵.

However, most of these approaches face two key shortcomings. First, the semantics of census units remain relatively static, typically adhering to the temporal granularity of the underlying census data. Second, while seldom discussed in the literature, they often focus on resident populations and overlook the influence of non-resident visitors on local characteristics. Consequently, these new census products fail to capture place dynamics and inter-relationships over time, remaining constrained by the same limitations as traditional censuses. In contrast, geospatial big data can address both challenges by providing more flexible and precise spatiotemporal profiles¹⁶. With individual movements at fine spatial and temporal scales, it reveals the dynamic relationships among different geographic units and can integrate this information into census-style data.

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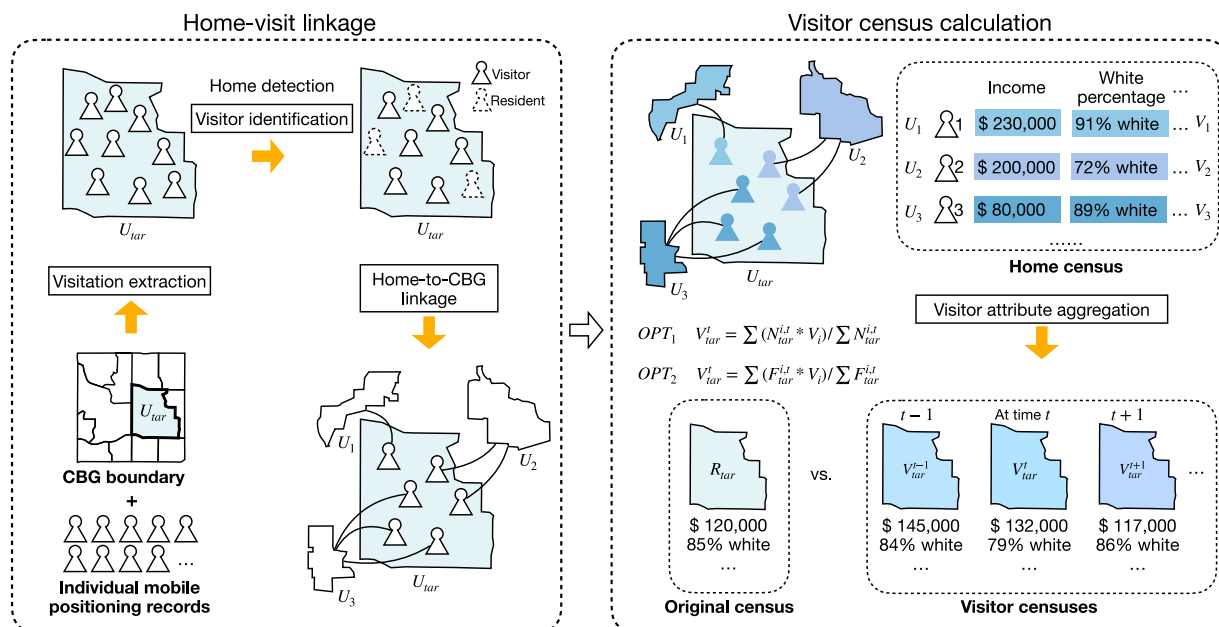


Fig. 1 Data generation framework of the visitor-enriched census: home-visit linkage and visitor census calculation. The former one contains visitation extraction, visitor identification (based on home detection), and home-to-CBG linkage, and the latter one focuses on visitor attribute aggregation. Ultimately, a series of visitor censuses during different time intervals (e.g., $t-1$, t , $t+1$) are generated and are compatible to the original resident-based census.

Although recent attempts to leverage geospatial big data have shown promise¹⁷, such efforts remain limited in both scope and diversity.

In this work, we present a dataset that can be viewed as a *visitor-enriched census* data product at the Census Block Group (CBG) level, incorporating mobile positioning data and the visitor perspective to meet the growing need for more detailed and dynamic census information. Specifically, for each CBG, individuals who visit but do not reside there are classified as visitors, and visitors' socio-demographic attributes – inferred by their home locations – enrich the CBG's semantics. By assigning the socio-demographic information of visitors to each census unit, this dataset represents a new type of resource characterized by dynamic census semantics derived from visitors rather than residents. Building on current census standards, the visitor-enriched census concept introduces an additional layer of information based on non-resident visitors within each geographic unit. Similar to origin-destination (OD) flow datasets, the intermediate *Home-Visit table* can also be visualized as flow patterns, thereby illustrating dynamics and interactions within a certain geographic area.

Methods

We generated visitor-enriched census tables by integrating information for both residents and visitors at the Census Block Group (CBG) level. We used the Unattributed Visits Dataset from PlaceIQ (<https://www.precisely.com/product/precisely-placeiq/placeiq-movement>) – an anonymized, multi-source commercial mobile positioning dataset – for extracting visitations, identifying visitors based on their home locations, and inferring spatiotemporal information for visitor census generation. This commercial dataset was provided under a non-exclusive, non-transferable license and complies with the California Consumer Privacy Act (CCPA). It captures users' locations each time they activate their devices, covering approximately 5% of the U.S. population and enables the tracing of individual trips and the inference of potential home locations. In this project, we focused on July 2021, as an example, to extract visitation patterns and derive visitors' socio-demographic attributes based on the census information of their presumed home CBG. To maintain temporal consistency, we applied 2017–2021 American Community Survey (ACS) five-year estimates provided by NHGIS (<https://www.nhgis.org/tabular-data-sources#acs>) for geographic unit matching and socio-demographic attribute extraction.

The data generation framework of our visitor census dataset consists of two major steps: *home-visit linkage* and *visitor census calculation*, as illustrated in Fig. 1.

Home-visit linkage. As illustrated (Fig. 1, left), *home-visit linkage* consists of three major steps. (1) Visitation extraction: At this step, we extracted all visitations to a target CBG, denoted as U_{tar} , based on individual mobile positioning records from PlaceIQ data. We first performed data cleaning and sorted all records by unique devices representing different individuals. For each device user, we collected a series of visitations, each containing fields including unique device ID, time, coordinate (lon-lat format), and stay duration. Then, we conducted spatial join to assign each visitation and its associated device to the corresponding target CBG. (2) Visitor identification: To classify an individual as a visitor or resident for each target CBG U_{tar} , we performed home location detection for all device users who visited the target CBG. For each visitation to U_{tar} , the individual was classified as a “visitor” if their

detected home location was outside U_{tar} . Otherwise, the individual was classified as a resident and excluded from the following visitor census generation processes. (3) Home-CBG linkage: Once visitors were identified based on home detection, the CBGs where visitors live, i.e., $\{U_1, U_2, U_3\}$, were linked to the target CBG U_{tar} to generate the visitor census attributes. For each target CBG, visitors' profiles were inferred by their identified potential home locations, linking the CBG they reside in (home) to the visited CBG (visit).

In detail, home location detection applied in the visitor identification step involves the following steps:

- **Nighttime record extraction.** We assumed that the location where an individual stays for the longest duration at nighttime to be the most possible home location¹⁸. Accordingly, we extracted records from 8 pm to 4 am referring to the previous studies¹⁹ and the observed hourly visitation distribution patterns in the PlaceIQ dataset. Then we selected device users with more than 20 nighttime records based on data density to exclude inactive individuals, which account for about 50% of individuals for each MSA, respectively.
- **Spatial clustering and home cluster inference.** Hierarchical Density-Based Spatial Clustering (HDBSCAN)²⁰ was employed to identify all dense spatial clusters formed by individual nighttime records, using pairwise Euclidean distances as a measure of spatial proximity. We defined the minimum size of each cluster as five, indicating that each cluster must comprise a minimum of five records. Here we consider the cluster with the highest average stay duration as the likely home location, as a higher stay duration in PlaceIQ data commonly represents no movements for a long time. For each individual, we selected the cluster with the highest average duration and tagged the cluster centroid as the home location. Note that the secondary homes were not considered for simplicity, despite that some individuals may have multiple homes in their daily life²¹.
- **Home location identification.** As censuses reflect the socio-demographic features by only residents²², we therefore exclude individuals who primarily spend nights in non-residential spaces. We applied a new building classification dataset covering the metropolitan areas in the entire U.S.²³, and conducted spatial join processes to define whether each estimated home location by clustering is within a residential area. In detail, home locations within a 20-meter buffer of buildings labeled as residential areas were selected to be the "real home locations" for visitor census generation. At the end of this step, socio-demographic information of individuals with validated home locations were input to generate the visitor census dataset.

Visitor census calculation. Home-visit CBG pairs of each target CBG U_{tar} generated during the *home-visit linkage* process were used as inputs for visitor census calculation. Socio-demographic attributes recorded in the census dataset, such as income and racial composition, can be aggregated spatially to produce estimates for visitor census.

Although methods such as multilevel statistical procedures²⁴ and the use of post-enumeration survey data²⁵ have been employed to aggregate or adjust traditional censuses, we do not incorporate them into our visitor census generation due to uniqueness of this census type and calculation complexity. Here two basic weighted aggregation methods are provided as example approaches to assign visitor information to each CBG: one based on the total number of visitors and another based on the total number of visits. We invite empirical demographic studies to enrich the visitor census aggregation methods from domain perspectives. Both methods listed below apply weights to CBGs, where the contributions of different visitor's home CBGs vary according to their interaction intensity weighted by different measures, rather than treating all CBGs equally as in simple average aggregation. Compared to visitor-based aggregation, visit-based aggregation emphasizes more on actual intensity of interaction than on individual profiles. Consequently, input features should be ratio variables (e.g., household income, percentage of white population), although categorical features (e.g., age groups, educational attainment categories) are also compatible if they can be expressed as ratios. The two basic options for visitor census calculation are explained as follows:

- **Option 1 - visitor-based.** Let V_{tar}^t represent any aggregated visitor feature for the target CBG within a specified time interval t . When the aggregation is based on the number of visitors, we define $N_{tar}^{i,t}$ as the number of visitors coming from CBG i who visited the target CBG U_{tar} within t . The aggregation can be formalized as:

$$V_{tar}^t = \frac{\sum(N_{tar}^{i,t} * V_i)}{\sum N_{tar}^{i,t}} \quad (1)$$

- **Option 2 - visit-based.** Similarly, if the aggregation is based on visits, let $F_{tar}^{i,t}$ denote the number of visits made by visitors who were estimated to be residing in CBG i during the selected time interval t . The aggregation is then:

$$V_{tar}^t = \frac{\sum(F_{tar}^{i,t} * V_i)}{\sum F_{tar}^{i,t}} \quad (2)$$

For both Eqs. (1) and (2), V_i refers to a ratio-based socio-demographic feature (e.g., average income, white percentage) of the sourcing visitor home CBG i . The time interval for measuring the number of visitors or visits is flexible, e.g., 10 minutes, an hour, a day, and an entire month, depending on the expected temporal granularity of the data.

Once the desired visitor census features have been obtained, the records can be extracted and stored in a series of tables. Because the visitor census shares the same CBG identifiers as the original census, it facilitates

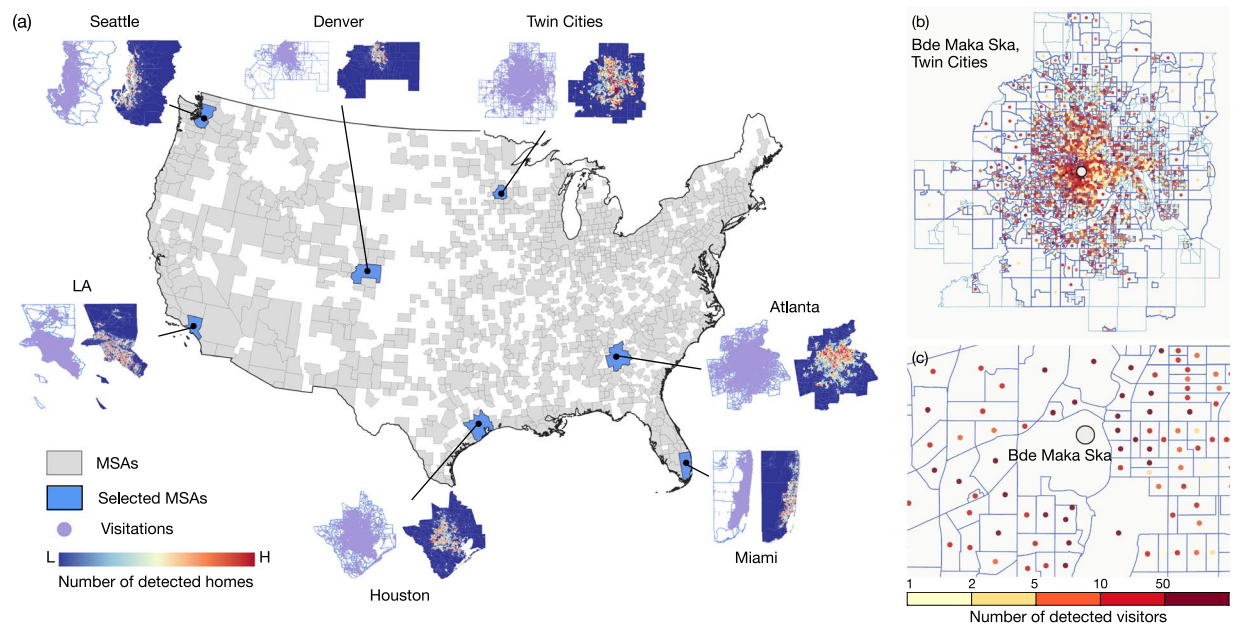


Fig. 2 Data illustration: from seven MSAs to a single CBG in the Twin Cities. **(a)** PlaceIQ dataset records (the lower left of each MSA name) along with home detection results aggregated at the CBG level (the lower right of each MSA name) in seven selected MSAs of the U.S. **(b)** An example of the Twin Cities: The numbers of visitors between a target CBG (Bde Maka Ska in Uptown Minneapolis) and other CBGs are rendered by colored dots, highlighting the connected CBGs. The darker the dot, the more visitors to the target CBG were detected. **(c)** A zoom-in view of the area near the target CBG.

direct comparisons between resident features (original census) and visitor features (visitor census) at the CBG level. In practice, the data generation framework can be adapted to different geographic units beyond the CBG level; in this paper, we simply demonstrate a CBG-level implementation as an example.

Visitor census datasets of seven MSAs. We select seven metropolitan statistical areas (MSAs) in different regions of the United States to showcase our datasets, including Los Angeles-Long Beach-Anaheim (LA), Houston-Pasadena-The Woodlands (Houston), Atlanta-Sandy Springs-Roswell (Atlanta), Miami-Fort Lauderdale-West Palm Beach (Miami), Seattle-Tacoma-Bellevue (Seattle), Denver-Aurora-Centennial (Denver), and Minneapolis-Saint. Paul (Twin Cities). As shown in Fig. 2a, the spatial distribution of visitations for each MSA (the left map under each MSA name) aligns broadly with the estimated home locations at the CBG level (the right map under each MSA name). Using the Twin Cities as an example, we select a CBG in Uptown Minneapolis to demonstrate how our dataset can be visualized as dot density maps (Fig. 2b,c). For simplicity, origin-destination (O-D) flows from visitors' home CBGs (visitor CBG) to the target CBG are omitted. The number of visitors, individuals who visit the target CBG but reside in other CBGs, provides the basis for this visitor census mapping. In the dot density maps, the total number of detected visitors from each visitor CBG with regard to the target CBG (Bde Maka Ska, Twin Cities, in this case) can be visualized, revealing the spatial heterogeneity of interaction intensity among each visitor CBG-target CBG pair. Such visitor census mapping can be generated for both the whole city (Fig. 2b) and any local area (Fig. 2c).

Building on the conceptual framework in Fig. 1, we provide a detailed illustration of how to generate *Visitor census table* from the original census dataset. Figure 3 highlights the table structure, field names, and the relationships among the tables. Specifically, the original census table - supplemented with spatiotemporal information from the PlaceIQ dataset - serves as the foundation for creating a *Home-Visit table* at the CBG level. This *Home-Visit table* adopts the unique CBG identifier (*CBG_ID*) assigned to each CBG and records the total number of visitors (*Num_visitor*) along with the total number of visits (*Num_visit*) from a visitor CBG (*Visitor_CBG*, short for visitor home CBG) to a target CBG (*Target_CBG*) for every hour (can be changed to other temporal granularities) in July 2021. Finally, for each CBG in the output *Visitor census table* (*CBG_ID_v*), it maintains the same structure as the original census table, while simply substituting the resident attributes (e.g., *Income*) with the aggregated visitor attributes (e.g., *Income_v*). In this study, we selected the following four economic and demographic attributes to exemplify the visitor-enriched censuses:

- The median household income (*Income* or *Income_v*) is the midpoint value of household incomes within a given population, reflecting the economic status of the households. It has also been widely employed to represent individual economic statuses^{26,27}.
- The white percentage (*Wp* or *Wp_v*) is the proportion of people identified as white in relation to the total population, relevant to the racial composition and social issues in certain regions.

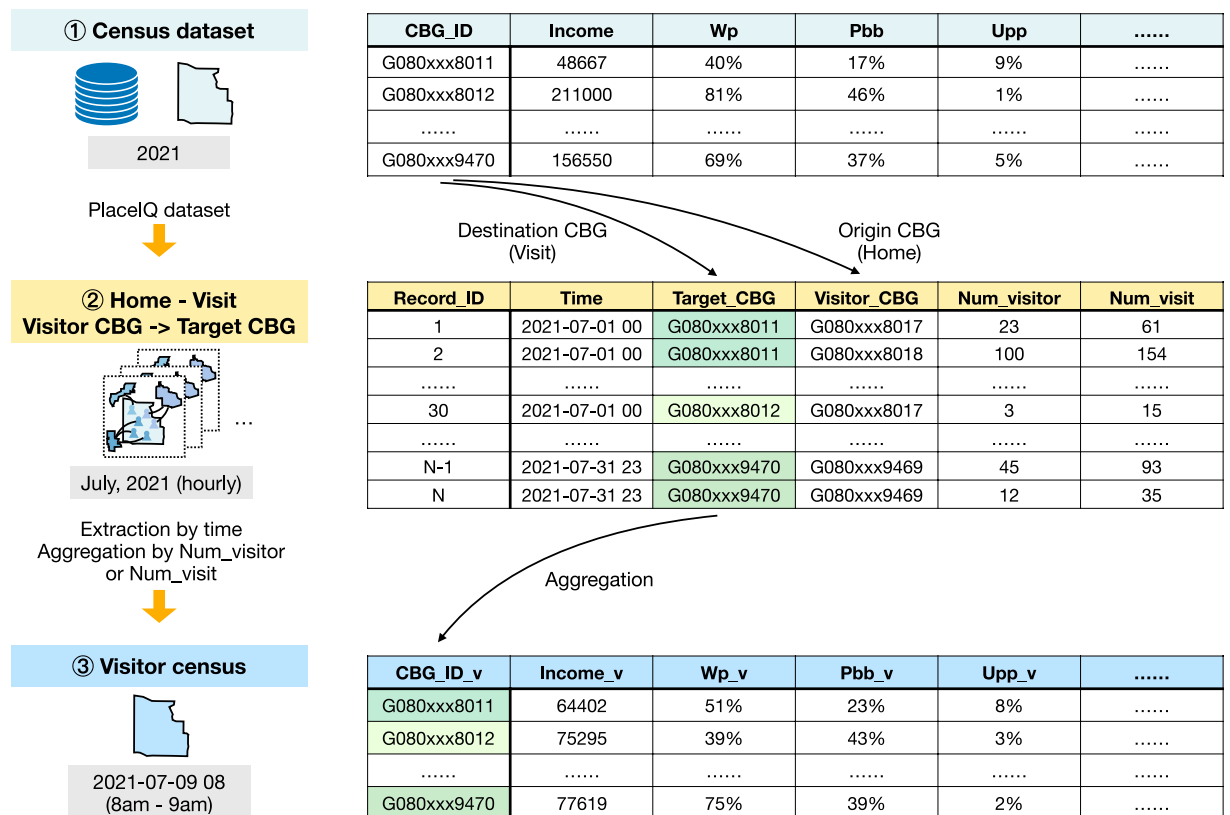


Fig. 3 An example of data processing from the census dataset (2021), the *Home-Visit table* (July 2021, hourly), to the *Visitor census table* (8 am - 9 am, July 9th 2021). Each pair of *Visitor_CBG* and *Target_CBG* in the *Home-Visit table* and *CBG_ID_v* in the *Visitor census table* adopt the unique CBG identifier (*GISJOIN*) from the census dataset.

- The percentage of individuals achieving bachelor's degree or better (*Pbb* or *Pbb_v*) refers to the proportion of people age 25 and older that have a bachelor's degree or higher within a given population, indicating the educational backgrounds of individuals²⁸.
- The unemployment percentage (*Upp* or *Upp_v*) represents the proportion of people age 16 and older that are unemployed within a given population, corresponding to the economic conditions and overall human well-being.

The comparisons between our visitor census (visitor-based aggregation) and the original 2017-2021 U.S. ACS records across seven MSAs are visualized in Fig. 4, demonstrating statistical distribution of four socio-demographic features across multiple cities. The feature value for a CBG is null if no visitors are detected throughout the entire month.

Data Records

The dataset is available in the figshare repository: <https://doi.org/10.6084/m9.figshare.28537322.v1>²⁹. For each MSA, we provide two versions of *Visitor census tables* (the third table in Fig. 3) based on two aggregation methods introduced in Eqs. (1) and (2), respectively. Consequently, there will be two CSV files per MSA (e.g., *Atlanta_visitor_July2021.csv* and *Atlanta_visit_July2021.csv* for Atlanta), documenting the aggregated visitor profiles over the entire month of July 2021. Users are recommended to link these tables to the original census data using the unique geographic identifier of each CBG - referred to as "*CBG_ID_v*" in the *Visitor census table* and "*GISJOIN*" in census tables or geographic boundary files at the CBG level, respectively.

In addition, we provide one-week (July 19-25, 2021) intermediate *Home-Visit tables* (the second table in Fig. 3) for each MSA (e.g., *Atlanta_homevisit_July21.csv.gz* for Atlanta on July 21st), which contain hourly visitor features instead of the monthly aggregates. Similar to the "*CBG_ID_v*" in visitor-enriched census tables, "*Target_CBG*" and "*Visitor_CBG*" serve as unique geographic identifiers for each CBG. These intermediate tables enable users to further customize and adapt the data for their specific needs, such as exploring alternative aggregation methods to highlight the relative contributions of different visitor CBGs.

Technical Validation

Given that there is no ground-truth information on device users' home locations, we adopted two validation methods based on common assumptions in human mobility research³⁰ to verify our home location estimation process (Fig. 5a) from the perspective of data source reliability (Validation one, Fig. 5b) and home location estimation accuracy (Validation two, Fig. 5c), respectively. In addition, we provide two real-world scenarios

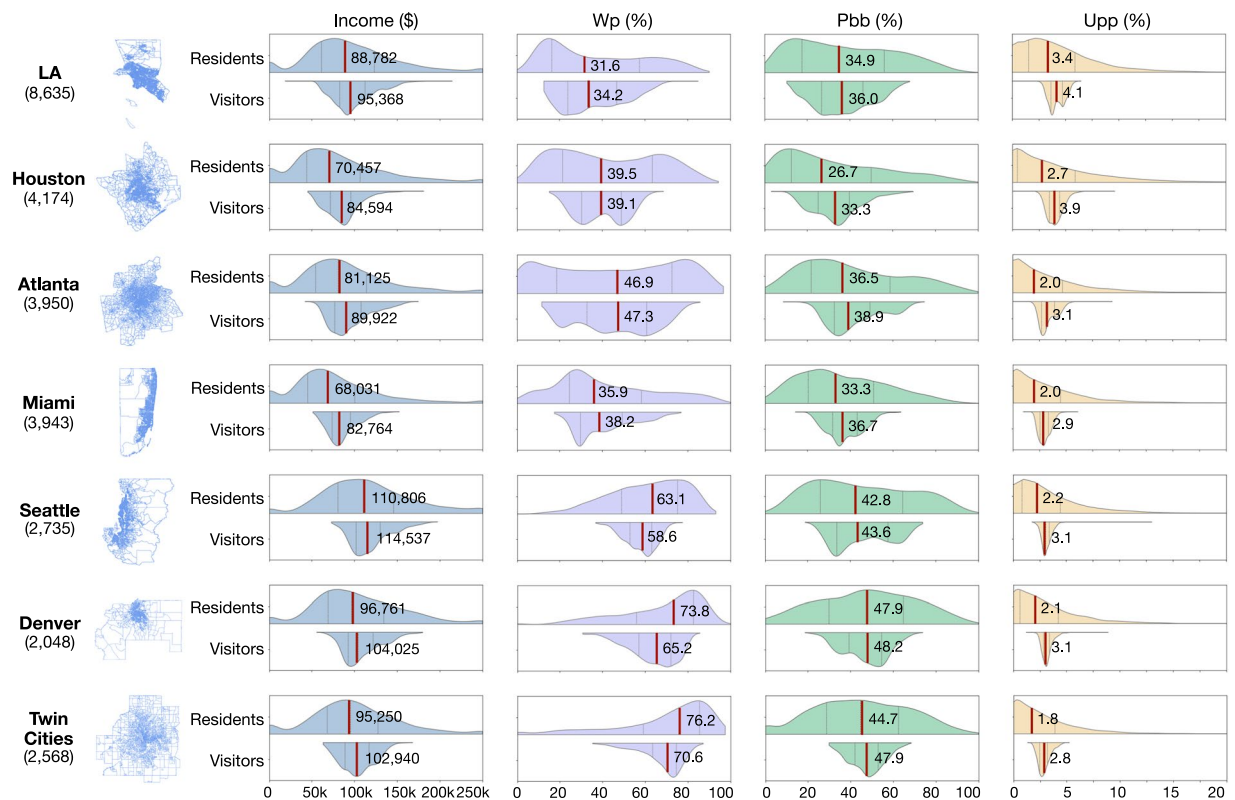


Fig. 4 Visitor census (visitor-based aggregation (Eq. (1)) based on July 2021 PlaceIQ) vs. original resident socio-demographics (2021 U.S. ACS). Displayed attributes include the median household income (*Income*), the white percentage (*Wp*), the percentage of individuals achieving bachelor's degree or better (*Pbb*), and the unemployment percentage (*Upp*). The number below each MSA name refers to the total number of CBGs.

examining the temporal dynamics (Validation three, Fig. 6) and spatial heterogeneity (Validation four, Fig. 7) of the visitor census dataset, respectively.

Validation one: home stay duration. The hourly average stay duration distribution of visitations in the whole month shows that individuals (Fig. 5b) are less likely to move during the nighttime³¹. All seven MSAs share similar temporal patterns, characterized by significantly longer average stay durations at night. Specifically, the average hourly stay duration during the daytime is approximately 4,247.39 seconds (~70 minutes), while at night it increases to approximately 8,637.94 seconds (~143 minutes). The pattern for Atlanta is slightly different but still follows the general trend. In other words, it is assumed that most individuals spend their nights at home as inferred from the PlaceIQ dataset, further verifying the applicability of one unified framework across multiple areas.

Validation two: population distribution. We applied the CBG-level population data by 2021 ACS records and compared it with the estimated home locations aggregated at the CBG level (Fig. 5c). As observed, the Pearson correlation coefficients between the census population and the number of visitors with detected homes at the CBG level are all above 0.4 (e.g., 0.56 for LA and 0.58 for Houston). Note that the coefficient values presented here may be underestimated by using the total population, as mobile positioning data covers primarily specific age groups and includes fewer young children and older adults³². In addition to biases related to age composition, the number of visitors with detected homes in the peripheral areas of the MSAs may be underestimated due to relatively sparse mobile records, leading to uncertainty in coefficient values. Besides, the sampling rates (number of visitors with detected homes versus the total population) of all MSAs are over 1% (e.g., 1.81% for LA and 2.78% for Miami) after home detection, which demonstrates the representativeness of the PlaceIQ dataset and the detected home locations.

Validation three: census dynamics. The dataset can reveal census dynamics by visitors at fine temporal scales based on hourly *Visitor census tables*. We output the hourly variation of non-white percentage in the Twin Cities. Fig. 6 depicts the non-white percentage disparity between nighttime and daytime in the Twin Cities reported by the visitor censuses (visitor-based aggregation). In detail, Fig. 6a plots the statistical and spatial distributions of the average differences in July, reflecting the overall temporal characteristics of the non-white percentage across the Twin Cities. To explore the local patterns, we picked two typical areas consisting of CBG clusters with higher non-white percentages at nighttime and daytime, respectively (Fig. 6b). The upper one is Downtown Minneapolis, an area where a significant number of non-white groups reside, despite frequent daytime visitations

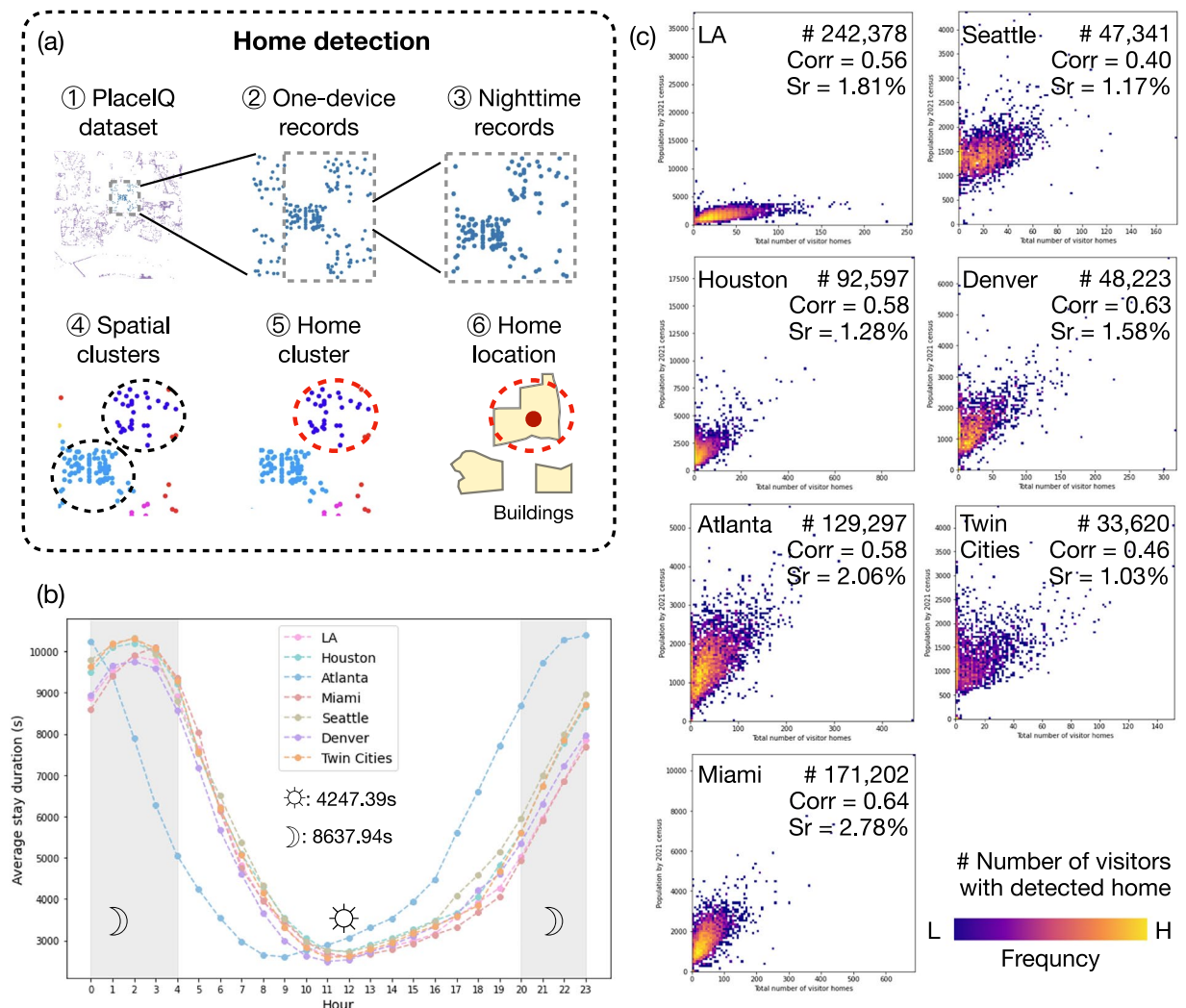


Fig. 5 Home detection processes and validation. **(a)** The flowchart of home location estimation. **(b)** The hourly distribution of average stay duration for each MSA. **(c)** The correlation between CBG-level population by 2021 ACS data and the total number of visitors with detected home along with Pearson correlation coefficient value (*Corr*) and sampling rate (*Sr*), after validation for each MSA.

by the working population coming from whiter communities. The lower one, Minnetrista City, is a residential area near the lakes, featuring a higher percentage of white residents. It is commonly visited by diverse social groups due to its recreational function, but reverts to an exclusive place for white residents at night. Intuitively, they represent two opposite examples of local racial fluctuations within a day. Further details are provided through two examples, illustrated by the average hourly non-white percentage of a single CBG inside each cluster (Fig. 6c). From the overall pattern to the single example, this scenario highlights the potential of our dataset in discovering certain social phenomena across various scales.

This example may further shed light on addressing specific socio-demographic phenomena rather than merely uncovering the general spatial and temporal patterns. One promising application is verifying the continued existence of sundown towns, places where non-white groups were historically forced to leave after sunset. Users are able to examine the disparities in non-white percentages today for places identified as sundown towns³³.

Validation four: socio-demographic heterogeneity. Apart from temporal variations, our visitor-enriched census dataset also captures the spatial heterogeneity of socio-demographic characteristics, offering valuable insights for socio-segregation research when compared to the traditional resident census. In this scenario, we calculated the disparity in median household income (Fig. 7a) and educational status (Fig. 7b) between residents and visitors in the Twin Cities. Similar to the last scenario, two CBG clusters are selected for each socio-demographic feature to demonstrate the typical patterns.

As for the income disparity, Fulton turns out to be a typical place where residents are significantly wealthier than visitors. The pattern is reversed for Downtown Saint Paul, an area characterized by visitors wealthier than residents. Apart from the working population, its amenities (e.g., bars and cafes) along the streets also attract

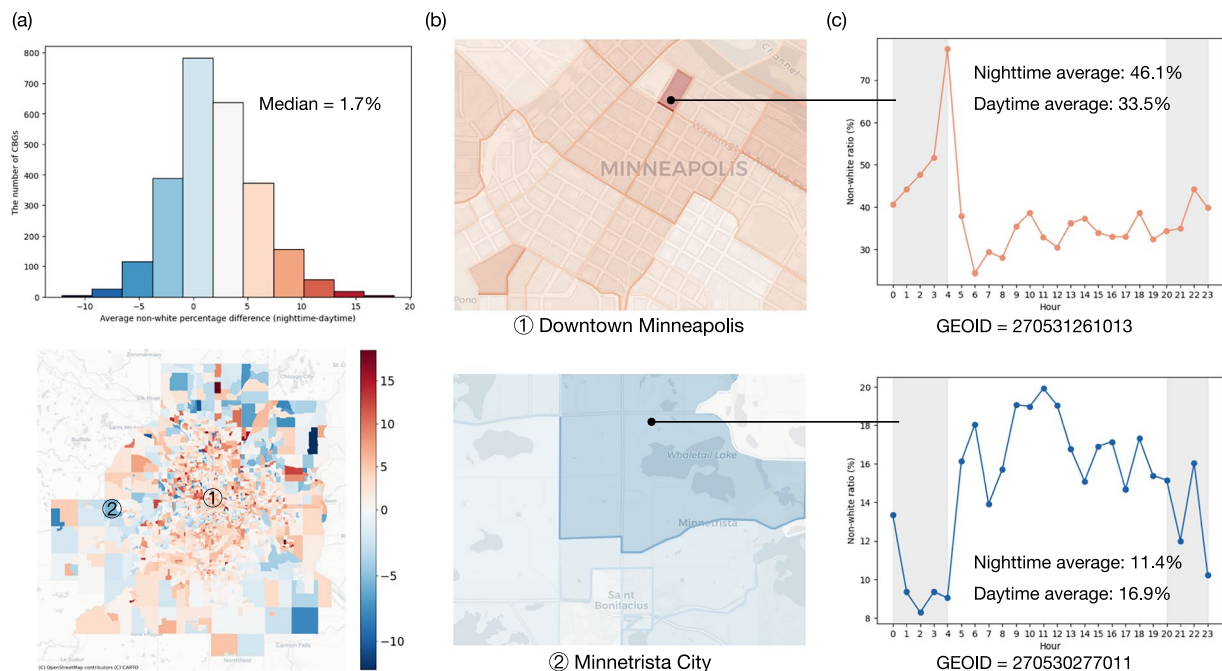


Fig. 6 The non-white percentage disparity between nighttime and daytime in July 2021, Twin Cities. **(a)** The overall disparity (nighttime average - daytime average) pattern across the whole Twin Cities. **(b)** Two typical areas of significantly higher non-white percentage during the daytime and nighttime respectively. **(c)** The average hourly non-white percentage change of two CBGs in **(b)** respectively.

frequent visitations by socially advantaged groups. An interesting thing is the spatial correlation between income level and educational level. Highly public areas such as Uptown Minneapolis, dotted with multiple lakes and recreational amenities, tend to have residents with higher levels of educational attainment compared to their highly diverse visitors. On the contrary, apartment-dense areas like Brooklyn Center serve as the home to socially disadvantaged groups but are frequently visited by individuals with higher educational levels. This is partly due to their location near major streets and the availability of services such as restaurants, grocery stores, and auto services.

Usage Notes

Our dataset can be applied in research and practice for purposes of extending the spatial and temporal dimensions of socio-demographics. It is recommended that users apply multiple exploratory spatial data analysis methods to examine and understand these socio-demographic disparities at both local and global levels. For instance, users are encouraged to calculate local Moran's I^{34} for each CBG, identify the significant clusters inside the city, and discover specific spatial patterns.

The visitor census dataset shares similarities with data products such as SafeGraph Social Distancing Metrics (<https://docs.safegraph.com/docs/social-distancing-metrics>) as we also provide OD-structured intermediate tables from visitors' potential homes to their destinations at the CBG level. Such hourly *Home-Visit tables* allow users to extract, aggregate, and analyze visitor semantics in a flexible manner. Therefore, our dataset delivers unique census semantics that set it apart from other OD data products and studies. In addition, compared to prevailing OD products or mobile positioning datasets (e.g., StreetLight O-D (<https://www.streetlightdata.com/origin-destination-od-study/>), HERE positioning (<https://www.here.com/platform/positioning/>)), ODs applied to generate our visitor census dataset delineate interactions only between locations where people reside and visit, rather than covering all trip purposes.

The generation of visitor census dataset can include different commercial or open positioning data as long as individual residential locations can be estimated, the PlaceIQ data is used as an example in this paper. Compared to other types of location-based data like social media data and GPS trajectories, mobile positioning data has two major advantages, rendering it more suitable for generating such a visitor census dataset. First, it offers both larger volume of data and greater representativeness across diverse social groups. Second, it captures a higher proportion of individual movements, as social media data tends to be more sparse in space and time and is not appropriate to support the detection of home locations³⁵. Unlike home detection and socio-demographic estimation conducted in many studies^{36,37}, our work is not intended to investigate the relationship between individual travel patterns and their socioeconomic status, nor to examine potential segregation; rather, it seeks to enrich the socio-demographic semantics of locations.

There are several concerns for this fusion dataset referring to its traits of being a census dataset and a mobile positioning derived dataset. First, like other socio-demographic estimation data products, our visitor census calculation is inevitably subject to ecological fallacy caused by individual socio-demographic profile estimation based on aggregated census data³⁸. In turn, users will encounter a similar issue when extracting aggregated

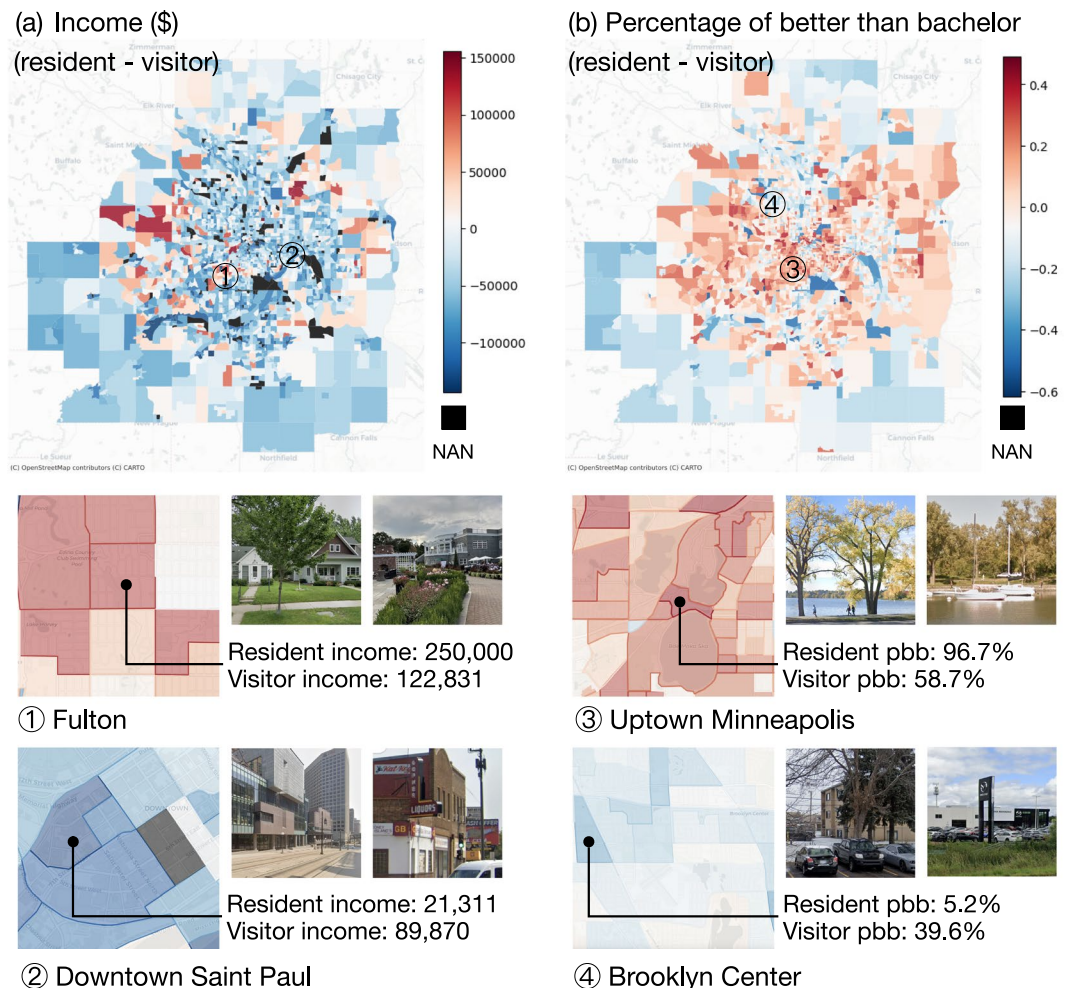


Fig. 7 The socio-demographic disparity between residents and visitors in July 2021, Twin Cities. **(a)** The income disparity (resident - visitor) pattern across the whole Twin Cities. **(b)** The educational disparity measured by the percentage of bachelor's degree or better.

socio-demographic features of visitors from this dataset. It turns out to be one of the notable challenges faced by demographic research and practice. Second, visitors included in this dataset are identified from mobile phone records, indicating data bias in group diversity. People who rarely use or lack access to mobile phones, such as young children, older adults, and those experiencing poverty, are underrepresented in this dataset³⁹. The third point is that we extracted visitation records inside the MSA boundaries, which means that we only consider visitors who live inside the corresponding MSAs, thus excluding visitors from external sources. In other words, it only reflects intra-city human interaction at neighboring scales.

Code availability

Data processing and data analysis were conducted using Python version 3.8. Code for dataset generation is available at https://github.com/GeoDI-Lab/Visitor_census. The open code is only designed for reproducing the visitor census calculation process as the original mobile-positioning data source is a commercial product. For additional details, users may directly contact the author team.

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Author contributions

D.Z. and M.X. conceived the study concept and design. D.Z. contributed to data acquisition. M.X. and D.Z. conducted the computational analysis. M.X. handled data processing and visualization. M.X. and D.Z. wrote the original draft. D.Z. and D.V.R. secured funding. M.X., D.Z., and D.V.R. jointly interpreted the data and revised the original draft.

Competing interests

The authors declare no competing interests.

Additional information

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